

A Pure Analytical Gaussian Expansion Method for Heavy and Mixed-State Quarkonia

PHYS400 Special Problems in Physics – Final Project Report

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Abstract

The investigation of heavy quarkonia acts as a crucial benchmark for the non-perturbative, low-energy regime of Quantum Chromodynamics (QCD). In this project, we systematically describe the mass spectra, wavefunctions, and decay properties of heavy mesons ($c\bar{c}$, $b\bar{b}$, $b\bar{c}$, $c\bar{u}$) using the phenomenological Cornell potential. Because the Coulomb-plus-linear structure of the Cornell potential lacks a purely analytical exact solution, we employ the Gaussian Expansion Method (GEM) combined with the Rayleigh-Ritz variational principle. This method translates the Schrödinger equation into a generalized eigenvalue problem, allowing for highly accurate, semi-analytical calculations of the energy spectrum.

We incorporated $\mathcal{O}(\alpha_s^2)$ relativistic corrections via the Breit-Fermi interaction to resolve fine and hyperfine splittings (spin-spin, spin-orbit, and tensor interactions). A complete computational pipeline was developed in Python, featuring a generalized eigenvalue solver, χ^2 parameter optimization against the 2024 Particle Data Group (PDG) data, and a robust error propagation module utilizing the finite difference method for mass uncertainties and decay widths (annihilation, radiative, and hadronic). The theoretical model demonstrates excellent agreement with experimental data, paving the way for extending the solver to three- and four-body multi-quark systems (baryons and tetraquarks) using Jacobi coordinates.

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1 Introduction

Hadrons containing heavy valence quarks, such as charmonium ($c\bar{c}$) and bottomonium ($b\bar{b}$), offer a unique window into the strong interaction governed by Quantum Chromodynamics (QCD). Due to the large mass of the charm ($m_c \approx 1.3$ GeV) and bottom ($m_b \approx 4.7$ GeV) quarks relative to the QCD scale parameter ($\Lambda_{\text{QCD}} \approx 200$ MeV), these systems are approximately non-relativistic. Consequently, they can be described with remarkable accuracy using non-relativistic potential models, similar to positronium in Quantum Electrodynamics (QED), albeit with the crucial addition of color confinement.

This research aims to comprehensively solve the heavy quarkonium system by optimizing the parameters of the Cornell potential to match the latest PDG 2024 mass spectra. We overcome the numerical limitations of standard finite-difference solvers by implementing the Gaussian Expansion Method (GEM). GEM is highly praised in few-body nuclear and hadronic physics for its rapid convergence and analytical integration capabilities. Furthermore, to provide a complete picture of quarkonium states, we compute vital observables, including root-mean-square (RMS) radii, wavefunctions at the origin, leptonic decay widths, and radiative transition rates, rigorously propagating errors from the χ^2 parameter fitting process.

2 Theoretical Framework

2.1 The Cornell Potential

The fundamental interaction between a heavy quark and antiquark is modeled using the Cornell potential, which linearly superposes a short-range Coulombic part (derived from one-gluon exchange) and a long-range linear confinement part (representing the QCD flux tube):

$$V_0(r) = -\frac{4}{3} \frac{\alpha_s}{r} + br + c \quad (1)$$

where α_s is the strong coupling constant, b is the string tension (confinement strength), and c is a constant mass shift encompassing the constituent quark masses and vacuum zero-point energy. The prefactor $4/3$ is the color factor for a color-singlet meson.

2.2 The Gaussian Expansion Method (GEM)

To solve the radial Schrödinger equation $H\Psi = E\Psi$, we apply the Rayleigh-Ritz variational method utilizing a basis of Gaussian functions. The radial wavefunction $R_{nl}(r)$ is expanded as:

$$R_{nl}(r) = \sum_{k=1}^N C_{nk} \phi_{kl}(r) \quad (2)$$

where the basis functions are chosen as Gaussians multiplied by the appropriate angular momentum barrier term:

$$\phi_{kl}(r) = N_{kl} r^l e^{-\nu_k r^2} \quad (3)$$

The normalization constant N_{kl} ensures $\int_0^\infty |\phi_{kl}(r)|^2 r^2 dr = 1$. A crucial advantage of GEM is the geometric progression of the Gaussian range parameters ν_k , which allows the basis to efficiently span both the highly localized short-range region and the diffuse long-range tails:

$$\nu_k = \nu_1 \left(\frac{\nu_N}{\nu_1} \right)^{\frac{k-1}{N-1}} \quad (4)$$

By projecting the Hamiltonian onto this non-orthogonal basis, the Schrödinger equation becomes a Generalized Eigenvalue Problem (GEVP):

$$\sum_{k=1}^N (H_{jk} - ES_{jk}) C_k = 0 \quad (5)$$

where $H_{jk} = \langle \phi_j | H | \phi_k \rangle$ is the Hamiltonian matrix element and $S_{jk} = \langle \phi_j | \phi_k \rangle$ is the overlap matrix. Because the basis functions are Gaussians, all matrix elements for kinetic energy, Coulombic potential, and linear potential can be computed entirely analytically using Gamma functions, drastically reducing computational overhead.

2.3 Relativistic Corrections: Breit-Fermi Hamiltonian

To accurately describe the physical meson spectrum, we must account for the fine and hyperfine structure, which arises from $\mathcal{O}(\alpha_s^2)$ and v^2/c^2 relativistic corrections. These are captured by the Breit-Fermi spin-dependent interactions treated as perturbations:

$$V_{\text{spin}}(r) = V_{SS}(r)(\mathbf{S}_1 \cdot \mathbf{S}_2) + V_{LS}(r)(\mathbf{L} \cdot \mathbf{S}) + V_T(r)S_{12} \quad (6)$$

where:

- **Spin-Spin (Hyperfine):** $V_{SS}(r) = \frac{32\pi\alpha_s}{9m_1m_2}\delta^3(\mathbf{r})$ splits the S-wave singlet (pseudoscalar, η_c) and triplet (vector, J/ψ) states.
- **Spin-Orbit (Fine):** $V_{LS}(r)$ splits the P-wave and D-wave triplets (e.g., $\chi_{c0}, \chi_{c1}, \chi_{c2}$) based on the total angular momentum J .
- **Tensor Interaction:** $V_T(r)$ mixes states of different orbital angular momentum but identical J (e.g., S-D mixing).

3 Computational Implementation

The project was entirely implemented in Python, modularized into distinct, maintainable sub-packages within the `src/quarkonia` directory.

3.1 Object-Oriented Design (`gem_solver.py`)

The core engine, `gem_solver.py`, dynamically constructs the Hamiltonian and Overlap matrices. Using `scipy.linalg.eigh`, it solves the GEVP to retrieve the energy eigenvalues E_n and the eigenvectors (the expansion coefficients C_{nk}). The output is strictly analytical down to the final matrix diagonalization, avoiding the discretization errors inherent in Runge-Kutta or Numerov methods.

3.2 Parameter Fitting (`fitter.py`)

To align the theoretical model with reality, `fitter.py` employs `scipy.optimize.minimize` to execute a χ^2 optimization of the parameter vector $\mathbf{p} = (\alpha_s, b, c, m_q, m_{\bar{q}})$. The loss function is defined as the Mean Squared Error (MSE) between the calculated theoretical masses and the experimental PDG 2024 dataset retrieved via `pdg_data.json`.

3.3 Decay Models & Observables (decay_models.py, observables.py)

The `observables.py` script calculates fundamental physical properties directly from the obtained GEM wavefunctions:

- **RMS Radius:** $\sqrt{\langle r^2 \rangle} = \sqrt{\int \Psi^*(r) r^2 \Psi(r) d^3r}$.
- **Wavefunction at Origin:** $|\Psi(0)|^2$, critical for evaluating S -wave decay processes.

The `decay_models.py` module implements transition dynamics:

- **Annihilation Decays:** Van Royen-Weisskopf formulas for leptonic decay rates $\Gamma(V \rightarrow e^+e^-)$.
- **Radiative Decays:** Computes Electric Dipole (E1) and Magnetic Dipole (M1) transitions between states (e.g., $\psi(2S) \rightarrow \chi_{cJ}\gamma$).
- **Hadronic Decays:** Implementing the 3P_0 quark-pair creation model for transitions above the open-flavor threshold.

3.4 Robust Error Propagation (metrics.py)

As identified during the late stages of development, determining the parametric uncertainty of the predictions is crucial. The covariance matrix from the χ^2 fit provides the variance in the fundamental parameters σ_p . To propagate these errors to complex non-linear observables (like M1/E1 transition widths), we implemented the finite difference method:

$$\sigma_f \approx \sqrt{\sum_i \left(\frac{f(p_i + \epsilon) - f(p_i - \epsilon)}{2\epsilon} \right)^2 \sigma_{p_i}^2} \quad (7)$$

with a step size of $\epsilon = 10^{-4}$. This guarantees robust confidence intervals for every predicted mass and decay width in the generated CSV reports.

4 Results and Discussion

4.1 Optimized Parameters and Mass Spectra

Through rigorous automated fitting over multiple sectors, the model found optimal parameters that successfully quench the MSE to the order of a few MeV per state. The optimization converged to the parameters listed in Table 1.

Table 1: Optimized Parameters for Heavy and Heavy-Light Quarkonia Sectors

Sector	α_s	b (GeV ²)	c (GeV)	m_q (GeV)	$m_{\bar{q}}$ (GeV)
Charmonium ($c\bar{c}$)	0.546	0.143	-0.250	1.48	1.48
Bottomonium ($b\bar{b}$)	0.358	0.189	-0.100	4.85	4.85
B_c Meson ($b\bar{c}$)	0.452	0.165	-0.175	4.85	1.48
D Meson ($c\bar{u}$)	0.612	0.130	-0.420	1.48	0.33

The spectra output by `run_spectrum.py` effectively reproduces the ground and excited states. The spin-dependent splitting successfully accounts for the mass gaps, such as between the $J/\psi(1S)$ vector state and the $\eta_c(1S)$ pseudoscalar state, directly driven by our analytical Breit-Fermi integrations.

Table 2: Calculated Mass Spectrum of Charmonium ($c\bar{c}$) States

State ($n^{2S+1}L_J$)	Meson	Calculated Mass (MeV)	PDG Mass (MeV)	ΔM (MeV)
1^1S_0	$\eta_c(1S)$	2981.5	2983.9	2.4
1^3S_1	$J/\psi(1S)$	3096.8	3096.9	0.1
1^3P_0	$\chi_{c0}(1P)$	3418.2	3414.7	3.5
1^3P_1	$\chi_{c1}(1P)$	3512.4	3510.6	1.8
1^3P_2	$\chi_{c2}(1P)$	3554.1	3556.1	2.0
1^1P_1	$h_c(1P)$	3526.8	3525.3	1.5
2^1S_0	$\eta_c(2S)$	3635.4	3637.5	2.1
2^3S_1	$\psi(2S)$	3687.2	3686.1	1.1
1^3D_1	$\psi(3770)$	3778.5	3773.7	4.8

Table 3: Calculated Mass Spectrum of Bottomonium ($b\bar{b}$) States

State ($n^{2S+1}L_J$)	Meson	Calculated Mass (MeV)	PDG Mass (MeV)	ΔM (MeV)
1^1S_0	$\eta_b(1S)$	9395.2	9398.7	3.5
1^3S_1	$\Upsilon(1S)$	9460.5	9460.3	0.2
1^3P_0	$\chi_{b0}(1P)$	9855.4	9859.4	3.3
1^3P_1	$\chi_{b1}(1P)$	9890.1	9892.7	2.6
1^3P_2	$\chi_{b2}(1P)$	9910.8	9912.2	1.4
2^3S_1	$\Upsilon(2S)$	10022.1	10023.2	1.1
2^3P_0	$\chi_{b0}(2P)$	10228.5	10232.5	4.0
2^3P_1	$\chi_{b1}(2P)$	10252.3	10255.4	3.1
2^3P_2	$\chi_{b2}(2P)$	10266.9	10268.6	1.7
3^3S_1	$\Upsilon(3S)$	10356.5	10355.2	1.3

4.2 Wavefunctions and Phenomenological Insights

By exporting the GEM coefficients to `[Sector]_Wave_GEM_Coefficients.csv`, we visualize the bound states. As detailed in Table 4, the model correctly exhibits spatial tightening (smaller RMS radii) for $b\bar{b}$ systems compared to $c\bar{c}$ systems due to the heavier mass and subsequently stronger Coulombic grip at short distances.

Table 4: Calculated Observables: RMS Radii, Wavefunction at Origin, and Leptonic Decay Widths

Meson	State	$\sqrt{\langle r^2 \rangle}$ (fm)	$ \Psi(0) ^2$ (GeV ³)	$\Gamma_{e^+e^-}$ (keV)
J/ψ	1S	0.412	0.0415	5.45 ± 0.12
$\psi(2S)$	2S	0.835	0.0201	2.21 ± 0.08
$\psi(3770)$	1D	1.025	~ 0	0.25 ± 0.04
$\Upsilon(1S)$	1S	0.218	0.4532	1.32 ± 0.03
$\Upsilon(2S)$	2S	0.495	0.2155	0.61 ± 0.02
$\Upsilon(3S)$	3S	0.732	0.1421	0.42 ± 0.02

4.3 Decay Widths and Error Margins

Thanks to the fully implemented finite-difference error propagation, the theoretical decay widths now feature rigorous uncertainty bounds (Table 5). The calculated M1 transition rates (e.g., $J/\psi \rightarrow \eta_c \gamma$) and E1 leptonic widths accurately mirror the experimental widths listed in the PDG, validating both the long-range confinement and the short-range wavefunction behavior ($\Psi(0)$) derived from the GEM solver.

Table 5: Radiative Decay Widths (E1 and M1 Transitions)

Transition	Type	Calculated Γ (keV)	Exp. PDG Γ (keV)
$J/\psi \rightarrow \eta_c \gamma$	M1	1.85 ± 0.15	1.58 ± 0.37
$\psi(2S) \rightarrow \eta_c(2S) \gamma$	M1	0.22 ± 0.05	0.21 ± 0.15
$\psi(2S) \rightarrow \chi_{c0} \gamma$	E1	28.5 ± 1.2	29.3 ± 1.2
$\psi(2S) \rightarrow \chi_{c1} \gamma$	E1	26.8 ± 1.1	28.7 ± 1.3
$\psi(2S) \rightarrow \chi_{c2} \gamma$	E1	19.4 ± 0.9	21.1 ± 0.9
$\Upsilon(2S) \rightarrow \chi_{b0} \gamma$	E1	1.15 ± 0.08	1.22 ± 0.16
$\Upsilon(2S) \rightarrow \chi_{b1} \gamma$	E1	2.10 ± 0.12	2.21 ± 0.22
$\Upsilon(2S) \rightarrow \chi_{b2} \gamma$	E1	2.32 ± 0.14	2.29 ± 0.22

5 Conclusion and Future Outlook

This project successfully constructed a high-precision, purely analytical solver for heavy quarkonia based on the Gaussian Expansion Method. The modular Python codebase not only extracts exact energy eigenvalues from the phenomenological Cornell potential but also rigorously fits parameters, evaluates complex radiative/hadronic decays, and propagates statistical uncertainties. The model proved extremely resilient, matching PDG 2024 benchmark data across the charmonium, bottomonium, and heavy-light (B_c , D) meson sectors.

Next Frontier (Future Work): With the 2-body ($q\bar{q}$) system fully mapped and operational, the immediate next step is extending the architecture to multi-quark exotic hadrons. Leveraging the flexibility of GEM, we will introduce Jacobi coordinates to transition the solver to 3-body

(baryons, e.g., Ω_{ccc}) and 4-body (tetraquarks, e.g., T_{cc}^+ , T_{bb}) systems. Because the basis functions simply become products of Gaussians over internal relative coordinates (e.g., ρ and λ), the analytical integration advantages of GEM will be preserved. To handle the exponential increase in basis size and matrix dimensionality, future iterations will port the computationally heavy tensor products and matrix diagonalizations into C++ utilizing the Eigen library, wrapped with a Python interface for user accessibility.

References

- [1] Particle Data Group (PDG), S. Navas *et al.*, *Review of Particle Physics*, 2024.
- [2] E. Hiyama, Y. Kino, M. Kamimura, *Gaussian expansion method for few-body systems*, Prog. Part. Nucl. Phys. 51 (2003) 223.
- [3] E. Eichten *et al.*, *Spectrum of Charmed Quark-Antiquark Bound States*, Phys. Rev. Lett. 34 (1975).